Argus-SQL: Finding the Accuracy–Latency Sweet Spot in Natural Language and Voice-to-SQL via Value-Grounded Foreign-Key Closure and Semantic Invariant Verification

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Clean-Room Open Research Framework (`argus-sql v0.1.0`)

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Abstract

Natural Language to SQL (NL2SQL) stands poised to become the foundational execution substrate of conversational and voice-driven user interfaces over enterprise-scale relational data. Yet a rigorous gap separates current academic leaderboard pipelines on benchmarks such as BIRD-SQL from interactive, voice-ready deployment: monolithic single-pass prompts collapse under schema overload and dirty cell literals, whereas brute-force multi-agent ensembles achieve high Execution Accuracy (EX) only by sampling dozens of homogeneous completions at prohibitive token cost and multi-second latency—while remaining vulnerable to correlated dialect hallucinations.

In this paper, we conduct a systematic compositional study of modern NL2SQL paradigms (`DIN-SQL`, `DAIL-SQL`, `MAC-SQL`, `MCS-SQL`, `CHESS`, and `CHASE-SQL`) and isolate five structural failure modes that govern real-world relational querying: (F1) spoken ASR disfluencies, lowercase multi-sentence drift, and colloquial quantifiers, (F2) dirty database cell-value drift and hierarchy shadowing, (F3) foreign-key bridge-table pruning collapse, (F4) silent SQLite dialect bugs (integer-division truncation, unfiltered `NULL` extrema, bare timestamp-to-date string comparisons, and missing window ranks), and (F5) naive majority-voting tyranny. By recombining orthogonal linking operators with deterministic post-execution verification, we propose **Argus-SQL**, a four-stage hybrid architecture uniting: (1) a *Voice & Intent Canonicalizer*, (2) an offline *Hierarchy-Aware Character 3-Gram Value Index paired with Foreign-Key Discounted Steiner Closure*, (3) *Complexity-Gated Diverse Batch Synthesis* ($K = 1$ fast-path routing for simple utterances vs. $K = 3$ orthogonal strategies for analytical queries), and (4) a *12-Invariant Semantic Verifier with Zero-Token Algebraic Dialect Repair and Lexicographic Execution Consensus*. To attribute accuracy gains without confounding LLM backbones or 20× multi-agent call budgets, we benchmark **Argus-SQL** against progressive, backbone-matched controlled operator baselines (`B0_RawDDL`, `B0+_EnrichedMDDL`, `B1_LexicalCoT`, `B2_ExecRefiner`, steelmanned `B3_ValueMajorityVote`, and an `MCS-SQL/CHESS`-style execution-preview `B4_LLMJudge`). Evaluated live with `gemini-3.7-flash` across 1,540 pipeline runs (7 systems × 220 queries) on an 11-database stratified `BIRD Mini-Dev` slice ($N = 110$ Clean and $N = 110$ spoken `Voice-BIRD` queries), **Argus-SQL** ranks **#1 of 7** on spoken `Voice-BIRD`—achieving 75.5% Voice EX (75.9 R-VES, 0.0% Clean-to-Voice degradation, 40.6% fewer tokens than `B3` on Voice) with a significant gain over raw-DDL (69.1%, exact McNemar $p = 0.039$) and higher EX than LLM-judge (72.7%), profiler-enriched full-DDL (73.6%), and steelmanned value majority-vote (74.5%). On Clean text, **Argus-SQL** (75.5% EX, 76.4 R-VES) trails steelmanned value majority-vote (77.3%) and LLM-judge (76.4%) by two and one queries, respectively (neither Clean difference significant at $N = 110$, exact McNemar $p \geq 0.27$ across all baselines). Metadata-enriched DDL alone lifts raw-DDL EX by +1.8 (Clean) and +4.5 (Voice) points. On our 48-query verification suite, **Argus-SQL** achieves 100.0% Clean and Voice EX, outperforming `B3_ValueMajorityVote` ($p = 5.96 \times 10^{-8}$) and beating `B4_LLMJudge` on spoken `Voice-BIRD` (100.0% vs. 70.8%, $p = 1.56 \times 10^{-2}$) at 18.7% lower token cost.

1 Introduction

The emergence of capable Large Language Models (LLMs) has catalyzed a fundamental shift in human–computer interaction: rather than navigating rigid dashboards or hand-authoring analytical queries, users increasingly expect to express intent directly in natural language—and, as voice becomes a primary ambient interface, through spontaneous spoken utterances. Within this paradigm, Natural Language to SQL (NL2SQL) occupies a uniquely high-leverage position. Unlike free-form retrieval-augmented generation, which struggles to aggregate millions of records faithfully, translating user intent into executable relational algebra allows a

conversational interface to query gigabytes or terabytes of structured data with mathematical precision.

Early cross-domain benchmarks such as Spider [13] demonstrated that neural models could master textbook relational syntax over clean, compact schemas. However, the introduction of the **BIRD-SQL** benchmark [6] exposed how severely textbook assumptions break down on realistic databases characterized by wide tables, noisy and abbreviated string cells, cryptic column identifiers, multi-hop foreign-key graphs, and domain-specific arithmetic formulas (*external evidence*). When spoken voice interfaces are layered atop such databases, Automatic Speech Recognition (ASR) disfluencies (*"uh," "you know"*), word stutters, spoken cardinals (*"top five"*), and colloquial shorthand (*"Calif," "LA county"*) compound schema and value ambiguity even further.

Recent leaderboard entries have pushed BIRD Execution Accuracy steadily upward through decomposed Chain-of-Thought prompting [12, 8], code-style DDL representations [3], multi-agent syntax refiners [10], locality-sensitive cell retrieval [9], and large-sample multi-path or multiple-choice ensembles [4, 5, 7]. Yet practitioners and reviewers confront two unresolved scientific questions:

1. **Controlled Operator Attribution**: When LLM backbone and candidate budget $K \in \{1, 3\}$ are held strictly constant, which specific linking, refinement, and selection operators actually drive gains across each failure mode—and where do popular operators silently fail?
2. **The Voice-Ready Sweet Spot**: Can sub-millisecond, zero-token semantic invariant verification and algebraic repair match or exceed an extra LLM Judge pass without incurring 20× sampling costs or succumbing to correlated dialect and spoken ASR errors?

Contributions. This work presents an empirical, engineering-grounded progression from baseline failure diagnosis to hybrid system synthesis:

- **Five-Mode Failure Taxonomy (§2)**: We formalize five recurring failure modes spanning spoken intent drift, dirty cell literals, broken foreign-key join subgraphs, silent SQLite arithmetic/`NULL` semantics, and majority-voting tyranny.
- **FK-Discounted Steiner Schema Linking (§3.2)**: We show that naive lexical linking suffers a dual pathology—exploding on star schemas via shared `*_id` keys while dropping lexically opaque bridge tables—and resolve both via surrogate-key seed discounting plus shortest-path foreign-key closure paired with hierarchy-aware character 3-gram value indexing.
- **Semantic Invariant Verification & Zero-Token Algebraic Repair (§3.4)**: Moving beyond syntax-only refiners [10, 1] and extra-call LLM judges [4, 9], **Argus-SQL** audits execution results against twelve dialect, intent, and relational invariants, deterministically repairing SQLite integer truncation across `SELECT` and `ORDER BY` clauses, guarding ascending `NULL` extrema, deduplicating 1-to-many join fanouts, rewriting display-name `GROUP BY` clauses onto unique join keys, and wrapping sub-day timestamp columns in `DATE(col)` in < 1 ms with zero extra LLM tokens.
- **Controlled Clean-Room Operator Suite (§4)**: Rather than conflating disparate 20-call multi-agent codebases, we implement six controlled, backbone-matched operator baselines (`B0_RawDDL`–`B4_LLMJudge`) alongside **Argus-SQL** and five leave-one-out ablations, paired with a SHA-256-verified split-environment evaluation harness.

2 Anatomy & Failure Taxonomy of NL2SQL

Table 1 synthesizes the design space of modern public BIRD-SQL systems. Rather than viewing these systems as isolated black boxes, we decompose an NL2SQL pipeline into four sequential operators over a database schema $\mathcal{S} = (\mathcal{T}, \mathcal{E}_{\text{FK}})$, database instance $\mathcal{D}$, external evidence e, and user utterance u:

$$\hat{q} = \mathcal{V}_{\text{arb}}\Big(\mathcal{G}_{\text{syn}}\big(\mathcal{L}_{\text{link}}(\mathcal{S}, \mathcal{D}, \mathcal{C}_{\text{int}}(u, e))\big),\ \mathcal{D}\Big), \qquad (1)$$

where $\mathcal{C}_{\text{int}}$ canonicalizes user intent, $\mathcal{L}_{\text{link}}$ grounds cell values and prunes the schema graph, $\mathcal{G}_{\text{syn}}$ synthesizes K SQL candidates, and $\mathcal{V}_{\text{arb}}$ executes, repairs, and arbitrates among candidates. Inspecting errors across existing instantiations of Eq. (1) reveals five orthogonal failure modes:

F1: Spoken Disfluency & Quantifier Drift. In voice-first interfaces, ASR transcripts contain conversational preambles (*"uh hey can you please tell me…"*), mid-clause fillers (*"you know"*), word stutters, and spoken number tokens (*"top three," "first five"*). Uncleaned transcripts pollute entity-span extraction and cause limit/complexity parsers to misclassify analytical queries.

F2: Dirty Database Cell-Value Drift. Users routinely utter abbreviations or colloquial variants (*"Calif," "LA county," "NY"*), whereas physical tables store normalized strings (`'California'`, `'Los Angeles'`, `'New York'`). Systems lacking a database cell index [8, 3, 10] emit syntactically valid predicates such as `WHERE state = 'Calif'` that silently evaluate to zero rows.

F3: Star-Schema Overload vs. Bridge-Table Collapse. Wide schemas containing legacy or staging tables (*e.g.*, `frpm_legacy` alongside `frpm`) confuse zero-shot full-DDL prompts (`B0`). Conversely, pure lexical

Table 1: Architectural decomposition and officially reported public BIRD-SQL Dev Execution Accuracy (EX %) of published multi-agent NL2SQL frameworks [2, 8, 3, 10, 4, 9, 7]. Published systems vary simultaneously in LLM backbone, few-shot retrieval, and 10–25-call agent budgets; in §4, we isolate their core linking, refinement, voting, and LLM-judge operators under a single controlled backbone ($K \in \{1, 3\}$).

Published System (Public Literature)	Voice Canon.	Cell Value Index	FK-Closure Pruning	Candidate Budget K	Refinement / Selection	Backbone (Reported)	Published BIRD Dev EX (%)
C3-SQL [2]	×	×	Lexical	$K = 10$–20	Naive Exec. Vote	GPT-3.5	50.2
DIN-SQL [8]	×	×	LLM Linker	$K = 1$ (4-Mod)	Generic Self-Corr.	GPT-4	50.7
DAIL-SQL [3]	×	×	Full DDL	$K = 1$–5	Masked k-Shot + Vote	GPT-4	54.8–55.9
MAC-SQL [10]	×	×	LLM Selector	$K = 1 + \text{Ref.}$	Syntax/Empty Refiner	GPT-4	59.4
MCS-SQL [4]	×	Prompted	Multi-Prompt	$K = 20$	LLM Multiple-Choice	GPT-4	63.4
CHESS [9]	×	LSH / MinHash	3-Step LLM	Multi-Unit	LLM Unit-Test Judge	GPT-4 / Open	65.0–68.3
CHASE-SQL [7]	×	LSH + Query	Multi-Path	$K = 21$	Pairwise Judge SFT	Gemini 1.5 Pro	73.0
Argus-SQL (Ours)	✓	**3-Gram Hier.**	**Steiner FK**	**Adaptive** 1/3	**12-Inv. + Alg. Repair**	**Model-Agnostic**	**See §4**

column pruning (`B1`) fails in two opposite ways: (i) if foreign-key columns such as `client_id` receive full lexical credit when the user mentions *"clients,"* every child table in a star schema is falsely retained; (ii) if intermediate junction tables carry lexically opaque names (*e.g.*, `account_disp` connecting `clients` to `loans`), lexical pruning drops the bridge table entirely, rendering a valid `JOIN` impossible.

F4: Silent SQLite Dialect & Formula Bugs. Execution-guided refiners such as `MAC-SQL` [10] trigger *only* when SQLite raises a runtime exception or returns an empty result set. On BIRD-SQL, however, two pervasive SQLite semantics produce non-empty, exception-free—yet completely wrong—answers:

1. *Integer-Division Truncation*: In SQLite, dividing two `INTEGER` expressions (`SUM(free_meals) / SUM(enrollment)`) truncates toward zero, returning `0` instead of `0.3125`—and if `ORDER BY a / b DESC LIMIT k` omits `REAL` casting, all ratios below 1.0 tie at `0`, corrupting top-k rankings.

2. *Unfiltered `NULL` Ascending Extrema*: SQLite treats `NULL` as smaller than any numeric value. Consequently, `ORDER BY sat_score ASC LIMIT 1` silently returns a row with `NULL` rather than the minimum recorded score whenever `WHERE sat_score IS NOT NULL` is omitted.

F5: Naive Majority-Voting Tyranny & Pure-Selector Limits. Self-consistency pipelines [11] select the largest execution equivalence class, actively overriding dialect-correct queries whenever $> 50\%$ of samples omit `CAST(... AS REAL)` or `IS NOT NULL`. Even when an extra LLM Judge pass [4, 9] inspects execution previews (`B4_LLMJudge`), any pure selector incurs a second sequential LLM round-trip and remains mathematically upper-bounded by Oracle@K whenever spoken ASR drift or correlated schema/timestamp traps cause $0/K$ candidates to succeed.

3 The Argus-SQL Architecture

Argus-SQL addresses F1–F5 by pairing lightweight deterministic pre- and post-processing with targeted, complexity-gated LLM synthesis.

3.1 Stage 1: Voice & Intent Canonicalization ($\mathcal{C}_{\text{int}}$)

Given raw text or ASR transcript u and evidence e, $\mathcal{C}_{\text{int}}$ deterministically strips filler interjections and conversational wake-phrases, collapses consecutive word stutters ($w_i w_i \to w_i$), normalizes spoken cardinals adjacent to ranking/comparison operators (*"top five"* $\to$ `top 5`), restores multi-sentence capitalization, terminal punctuation, and canonical Proper-Noun/identifier casing (*e.g.*, `TR004_8_9`, `Marvel Comics`, `DC`) attested in e, extracts candidate entity and alphanumeric code spans $E(u)$, estimates expected projection arity $\hat{a}$ across multi-sentence follow-up directives (`? Show...`, `? Indicate...`) and imperative window rankings (`Rank` X `by` $Y \to 3$) alongside strict single-column formula locks (`calculation =`), and assigns a complexity tier $\tau(u, e) \in \{\textsc{Simple}, \textsc{Moderate}, \textsc{Challenging}\}$.

3.2 Stage 2: 3-Gram Value Index & FK-Discounted Steiner Linking ($\mathcal{L}_{\text{link}}$)

Offline Hierarchy-Aware Character 3-Gram Index. To ground entity mentions without issuing per-query `LIKE` scans across large tables, we index distinct non-null `TEXT` literals and per-column distinct cardinalities once per database (bounded by $M_{\max} = 500$ distinct cells per column via a read-only URI connection). Each normalized cell value v is mapped to boundary-padded character trigrams $G_3(v) = \{v'[i : i + 3]\}$ in an inverted posting list. For each query span $s \in E(u)$, candidate cells sharing at least one trigram are scored by blending

Jaccard overlap with Gestalt sequence similarity:

$$\mathrm{Sim}(s,v)=\begin{cases}1.0, & \tilde{s}=\tilde{v},\\ \max\big(0.86,\rho_{\mathrm{seq}}(\tilde{s},\tilde{v})\big), & \mathrm{pref}(\tilde{s},\tilde{v}),\\ 0.45J_3(\tilde{s},\tilde{v})+0.55\rho_{\mathrm{seq}}(\tilde{s},\tilde{v}), & \text{else},\end{cases} \tag{2}$$

where $J_3(\tilde{s},\tilde{v})=|G_3(\tilde{s})\cap G_3(\tilde{v})|/|G_3(\tilde{s})\cup G_3(\tilde{v})|$. To prevent false-positive prefix pollution and geographic shadowing, $\mathcal{L}_{\mathrm{link}}$ blocks non-exact prefix matches on schema identifier tokens and parenthetical column qualifiers (*e.g.*, `(K-12)`), drops unmentioned `Mail*` mirror duplicates, and suppresses high-cardinality column shadows (≥ 250 distinct cells, such as `City`) whenever the identical literal matches a canonical low-cardinality hierarchy dimension (≤ 120 distinct cells, such as 58-value `County`) and the high-cardinality column noun is absent from the query. Surviving matches are injected into DDL column comments (`-- matched_values: ['California']`) while retaining distinct domain `examples:` on undocumented columns [3, 9].

FK-Discounted Seed Selection & Junction Closure. For each table $T \in \mathcal{T}$ and column $c \in T$, we compute a composite relevance score:

$$S(T,c)=S_{\mathrm{lex}}(u,T,c)+1.35\,S_{\mathrm{ev}}(e,c)+1.65\,S_{\mathrm{val}}(T,c), \tag{3}$$

where S_{lex} applies morphological, localization, and compound sub-word tokenization (*e.g.*, splitting `gasstations` into `gas` + `station` and bridging `translation` to `foreign`/`language`) over column names and merged CSV value codebooks. Crucially, to prevent star-schema seed explosion (F3), if c is a primary key or surrogate foreign key (`*_id`) on a table T whose own name has zero lexical overlap with u and e, we apply a $0.25\times$ seed discount to S_{lex}. Furthermore, when an isolated orphan table ($d_{\mathrm{FK}}(T)=0$, such as a legacy backup table) matches only a subset of columns already covered by an FK-connected seed table, the orphan distractor is pruned. Finally, augmenting declared foreign keys with exact single-column entity-PK matches on schemas that omit DDL constraints, we compute all-pairs shortest paths over the undirected foreign-key graph $G_{\mathrm{FK}}=(\mathcal{T},\mathcal{E}_{\mathrm{FK}})$ via Breadth-First Search and include many-to-many association tables referencing ≥ 2 selected tables plus compact ($\leq$ 16-column) lookup parents, retaining 100% of columns on narrow and medium tables (≤ 50 columns—achieving 99.8% offline / 100.0% value-grounded table recall and 100.0% selected-table column recall across all 500 `BIRD Mini-Dev` queries) and rendering composite keys as table-level `PRIMARY KEY` (c_1,c_2) constraints alongside batched read-only `MIN..MAX` temporal and numeric domain bounds (*e.g.*, `range: '2012-08-23'..'2012-08-26'`, `range: 0.0018..1.0`).

3.3 Stage 3: Complexity-Gated Batch Synthesis ($\mathcal{G}_{\mathbf{syn}}$)

Uniformly sampling $K=20$ completions per question wastes latency and tokens on straightforward single-table lookups. **Argus-SQL** routes queries dynamically:

- **Voice Fast-Path ($K=1$)**: When $\tau(u,e)=$ SIMPLE and $e=\emptyset$, a single deterministic `formula_defensive` prompt is dispatched, preserving sub-second voice responsiveness.
- **Diverse Multi-Strategy Batch ($K=3$)**: For MODERATE and CHALLENGING queries, three structurally distinct prompts—`decomposed_cot` (stepwise join parsimony, entity-key grouping, per-entity count-subquery, and range planning), `formula_defensive` (explicit SQLite cast, `INNER JOIN + CASE WHEN` ratios, `DATE(col)` timestamp guards, window ranks, and `NULL`-sort rules), and `skeleton_minimal` (projection-minimal direct relational anchor)—are dispatched *concurrently* in a single deduplicated batch request.

3.4 Stage 4: Semantic Invariant Verification, Algebraic Repair & Consensus ($\mathcal{V}_{\mathbf{arb}}$)

Hardened Read-Only SQLite Sandbox. Every unique SQL candidate is executed once inside a security-hardened SQLite sandbox opened via RFC 8089 `file:///<db>?mode=ro`, locked by a C-level `sqlite3_set_authorizer` allow-listing exclusively `SQLITE_SELECT`, `SQLITE_READ`, and safe scalar/aggregate functions (hard-denying `PRAGMA`, `ATTACH`, `DROP`, `DELETE`, `load_extension`, and blob memory bombs `zeroblob`/`randomblob`), and bounded by a virtual-machine opcode budget via `set_progress_handler` so runaway cross-joins abort cleanly.

Twelve Dialect & Relational Semantic Invariants. For each candidate q_i and execution outcome o_i, our verifier checks: (I1) `EXECUTION_ERROR`, (I2) `EMPTY_RESULT_ANOMALY` ($|o_i|=0$, surfacing top grounded literals and visible DDL `range:` bounds), (I3) `ALL_NULL_RESULT_ANOMALY` ($o_i=[(\texttt{NULL},\dots)]$), (I4) `INTEGER_DIVISION_TRUNCATION` (division `/` under ratio intent without `CAST(... AS REAL)`, `AVG`/`TOTAL`, or float-literal promotion), (I5) `UNFILTERED_NULL_ASC_LIMIT` (`ORDER BY col [ASC] LIMIT k` without `col IS NOT NULL`), (I6) `PROJECTION_ARITY_DRIFT` ($|\mathrm{cols}(o_i)-\hat{a}|\geq 2$, or $\mathrm{cols}(o_i)\neq 1$ on strict scalar `calculation =` formulas), (I7) `JOIN_FANOUT_DUPLICATES` (multi-table `JOIN` entity listings returning $\geq 50\%$ duplicate rows without `DISTINCT`), (I8) `LEFT_JOIN_FILTERED_RATIO` (filtering a 1-to-many non-PK child table inside `LEFT JOIN ... ON ... AND pred` while computing un-deduplicated

COUNT ratios), (I9) NON_KEY_ENTITY_GROUPING (grouping equi-joined entity tables by display names such as GROUP BY T.team_long_name instead of their unique join key), (I10) BOOLEAN_AVG_FOR_COUNT (evaluating SQLite 0/1 boolean AVG(col = 'x') proportions on average-count-per-entity tasks), (I11) TIMESTAMP_BARE_DATE_COMPARISON (comparing sub-day ISO timestamp columns whose DDL range: contains HH:MM bare against 'YYYY-MM-DD' date literals via =, >, <=, BETWEEN, or IN, which misclassifies same-day timestamps under ASCII string order), and (I12) MISSING_WINDOW_RANK (omitting RANK() OVER (ORDER BY ...) on imperative Rank X by Y queries).

Zero-Token Algebraic Dialect & Relational Repair. Before invoking an expensive LLM repair pass, **Argus-SQL** applies literal-preserving algebraic rewrites to any candidate triggering I1–I5, I7, I9, or I11: rewriting non-SQLite ILIKE operators, promoting uncast column, aggregate, or parenthesized arithmetic divisions lhs / rhs across SELECT and ORDER BY clauses to CAST(lhs AS REAL) / NULLIF(rhs, 0) (while preserving slashes inside quoted literals such as 'Free/Reduced'), injecting outer-scope col IS NOT NULL guards prior to ascending LIMIT sorts, deduplicating 1-to-many join fanouts via SELECT DISTINCT, rewriting display-name GROUP BY clauses onto their unique equi-join entity ID key, wrapping sub-day timestamp columns in DATE(col) on bare 'YYYY-MM-DD' predicates, and substituting abbreviated or mis-cased grounded cell literals (Sim ≥ 0.75) on empty or all-NULL results. Only if zero error-free candidates exist after algebraic repair (including I8, I10, or I12 errors) does **Argus-SQL** issue a single diagnostic LLM repair call (supplying DDL range: bounds and sibling failed SQL attempts), hardening the repaired query once more through zero-token algebraic rewrites.

Lexicographic Invariant-Weighted Consensus. Let $\mathcal{C}_r = \{i : \text{sig}(o_i) = r\}$ denote the equivalence cluster of candidates sharing normalized set/ordered result SHA-256 digest r. Each candidate receives weight $w_i = \gamma_{\text{err}}^{n_{\text{err}}(i)} \gamma_{\text{warn}}^{n_{\text{warn}}(i)}$ ($\gamma_{\text{err}} = 0.18, \gamma_{\text{warn}} = 0.48$, or 0 if execution failed), and clusters are ranked lexicographically by $(\mathbb{I}[\exists i \in \mathcal{C}_r : n_{\text{err}}(i) = 0], \text{Score}(\mathcal{C}_r))$, where:

$$\text{Score}(\mathcal{C}_r) = \sum_{i \in \mathcal{C}_r} w_i + \beta_{\text{div}} \max\big(0,\, |\mathcal{S}_{\text{clean}}(\mathcal{C}_r)| - 1\big), \quad (4)$$

with $\beta_{\text{div}} = 0.15$ rewarding strategy diversity $\mathcal{S}_{\text{clean}}(\mathcal{C}_r)$ strictly across *completely issue-free* cluster members. Under Eq. (4), error-only clusters (0-row, all-NULL, uncast truncation, bare timestamp mis-filtering, or 1-to-many LEFT JOIN ratio distortion) can never outvote an error-free cluster at any ensemble size K, and a single invariant-clean candidate (1.00) strictly beats two warning-flagged candidates ($2 \times 0.48 = 0.96$).

4 Experimental Evaluation

4.1 Experimental Setup & Controlled Operator Baselines

We evaluate all systems using two official BIRD-SQL metrics [6] plus paired exact significance testing: (1) **Execution Accuracy (EX %)**, measuring relational set equivalence (or row-order-preserving equivalence when ORDER BY is present) between predicted and gold result sets with IEEE-754 floating-point tolerance; (2) **Reward-based Valid Efficiency Score (R-VES)**, applying the official BIRD Mini-Dev piecewise speed-ratio reward ($\{0, 0.25, 0.50, 0.75, 1.00, 1.25\}$ scaled to 0–100) over median interleaved warm sandbox runs; and (3) two-sided **exact paired McNemar binomial p-values** on discordant predictions.

Controlled Operator Baselines vs. Multi-Agent Literature Systems. While Table 1 reports official numbers from full third-party multi-agent systems (*e.g.*, DIN-SQL's 4-module few-shot pipeline, DAIL-SQL's masked training-set k-shot retrieval, and MCS-SQL/CHESS/CHASE-SQL's 10–25-call agent ensembles), porting full multi-agent pipelines conflates 10×–20× token-budget disparities with individual operator efficacy. To isolate each architectural mechanism fairly under an identical LLM port, prompt protocol, and read-only sandbox ($K \in \{1, 3\}$), we construct six progressive **Controlled Operator Baselines**:

- B0_RawDDL: True un-enriched CREATE TABLE DDL (column types and PK/FK constraints only; zero inline metadata comments), $K = 1$.
- B0+_EnrichedMDDL (legacy ID B0_ZeroShotDDL): Full unpruned schema enriched with our offline statistical MIN..MAX range:, SQLite examples:, and merged CSV codebook annotations, $K = 1$—cleanly isolating offline M-Schema profiling from B0_RawDDL.
- B1_LexicalCoT (legacy ID B1_DINDecomposed): Deterministic lexical column pruning (no cell index or FK closure) + single-pass decomposed CoT ($K = 1$).
- B2_ExecRefiner (legacy ID B2_MACRefiner): FK-closed schema linking + bounded SQLite syntax/empty-result LLM refinement (1 repair pass max).
- B3_ValueMajorityVote: Steelmanned character 3-gram cell-value grounding (with fair multi-word proper-noun, quoted-literal, and code extraction) + FK-closed linking + $K = 3$ unweighted execution self-consistency voting.
- B4_LLMJudge: Identical steelmanned value-grounded $K = 3$ candidate pool as B3, augmented with an MCS-SQL/CHESS-style Multiple-Choice & Execution-Preview LLM Judge invoked whenever executing candidates form ≥ 2 distinct result clusters.

Table 2: Controlled diagnostic evaluation on Mini-BIRD (24 clean relational queries) and Voice-BIRD (24 spoken ASR-perturbed utterances) across controlled backbone-matched operator baselines (B0–B4) and Argus-SQL.

System / Configuration	Simple EX	Mod. EX	Chall. EX	Clean EX (%)	Clean R-VES	Voice EX (%)	Mean Tokens
B0_RawDDL	75.0	37.5	0.0	37.5	37.5	16.7	553
B0+_EnrichedMDDL	75.0	37.5	0.0	37.5	37.5	16.7	568
B1_LexicalCoT	100.0	25.0	0.0	41.7	41.7	25.0	676
B2_ExecRefiner	100.0	37.5	0.0	45.8	45.8	25.0	677
B3_ValueMajorityVote	100.0	62.5	0.0	54.2	54.2	41.7	1984
B4_LLMJudge	100.0	100.0	100.0	100.0	100.0	70.8	2204
Argus-SQL	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	1792

Table 3: Leave-one-out component ablation study of Argus-SQL across Clean and Voice-BIRD diagnostic splits, isolating each module's impact on Execution Accuracy (EX), R-VES, and prompt token budget.

System / Configuration	Simple EX	Mod. EX	Chall. EX	Clean EX (%)	Clean R-VES	Voice EX (%)	Mean Tokens
Argus-SQL	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	1792
w/o Voice Canonicalizer	100.0	100.0	87.5	95.8	95.8	70.8	1966
w/o 3-Gram Value Grounding	100.0	75.0	50.0	75.0	75.0	62.5	1779
w/o Steiner FK Closure	100.0	62.5	50.0	70.8	70.8	70.8	1938
w/o Semantic Verifier & Repair	100.0	62.5	0.0	54.2	54.2	54.2	1792
w/o Adaptive Fast-Path (Fixed K=3)	100.0	100.0	100.0	100.0	100.0	100.0	1984

Controlled Diagnostic Verification (`Mini-BIRD` & `Voice-BIRD`). To verify operator behavior deterministically without API stochasticity, we materialize three multi-table SQLite databases (`schools_mini`, `finance_mini`, and `clinical_mini`), each containing a parent entity table, a lexically opaque FK bridge table, a fact table with nullable scores and integer ratio columns, and a legacy orphan distractor table, across 24 stratified clean queries (8 Simple, 8 Moderate, 8 Challenging) and 24 paired **Voice-BIRD** spoken utterances (`SchemaGuidedDiagnosticSynthesizer`).

4.2 Comparative Progression & Ablation Analysis

Table 2 traces the operator progression from `B0_RawDDL` to **Argus-SQL**:

1. **Schema Overload vs. Lexical & FK-Refiner Operators (`B0/B0+` → `B2`)**: Both `B0_RawDDL` and `B0+_EnrichedMDDL` stall at 37.5% Clean EX because unpruned legacy tables (`frpm_legacy`, `loans_legacy`) hijack table-agnostic aggregations. `B1_LexicalCoT` eliminates orphan distractors on simple lookups (100.0% Simple EX) but drops intermediate bridge tables (25.0% Mod. EX); `B2_ExecRefiner` restores multi-hop FK joins (45.8% Clean EX) yet remains blind to silent SQLite integer truncation and `NULL` extrema.

2. **Steelmanned Majority Vote vs. Execution Judge (`B3` → `B4`)**: `B3_ValueMajorityVote` grounds multi-word and prefix literals (62.5% Mod. EX, 54.2% Clean EX), but 2/3 non-defensive strategies outvote the single `REAL`-cast candidate (0.0% Chall. EX). Adding `B4_LLMJudge` rescues all Clean `NULL`-sort and division disagreements (100.0% Clean EX)—confirming `B4` as a strong steelman selector—yet requires +23.0% more tokens than **Argus-SQL** (2,204 vs. 1,792) plus a second sequential LLM round-trip, and drops −29.2 points on spoken `Voice-BIRD` (70.8% EX) whenever spoken quantifiers or colloquial shorthands cause 0/3 candidates to succeed (Oracle@3 = 0%).

3. **The Argus-SQL Sweet Spot**: Uniting voice canonicalization, FK-discounted Steiner linking, 3-gram grounding, and < 1 ms zero-token algebraic repair achieves 100.0% Clean and Voice EX ($p = 5.96 \times 10^{-8}$ vs. `B3_ValueMajorityVote`; $p = 1.56 \times 10^{-2}$ vs. `B4_LLMJudge`) at 1,792 mean tokens (−9.7% vs. `B3`, −18.7% vs. `B4`).

Table 3 confirms via leave-one-out ablations that every stage is load-bearing: disabling Semantic Verification & Algebraic Repair causes the sharpest drop (−45.8 EX points), followed by removing Steiner FK Closure (−29.2 points), 3-Gram Value Grounding (−25.0 Clean / −37.5 Voice points), and Voice Canonicalization (−4.2 Clean / −29.2 Voice points), while disabling the Adaptive Fast-Path inflates mean token consumption (1,984 vs. 1,792) with zero accuracy benefit.

4.3 Live Public BIRD Mini-Dev Evaluation

Table 4 reports live multi-database evaluation over the official 11-database public `BIRD Mini-Dev` SQLite benchmark using `gemini-3.7-flash` via SHA-256-verified split-environment telemetry bundles ($N = 110$ difficulty-stratified Clean tasks and $N = 110$ paired `Voice-BIRD` utterances across 10 tasks per database—37 Simple, 37 Moderate, and 36 Challenging, totaling 1,540 end-to-end pipeline evaluations across the 7-system con-

Table 4: Live BIRD-SQL evaluation using gemini-3.7-flash.

System / Configuration	Simple EX	Mod. EX	Chall. EX	Clean EX (%)	Clean R-VES	Voice EX (%)	Mean Tokens
B0_RawDDL	81.1	73.0	58.3	70.9	71.6	69.1	1926
B0+_EnrichedMDDL	81.1	75.7	61.1	72.7	73.0	73.6	3647
B1_LexicalCoT	83.8	75.7	58.3	72.7	73.6	70.9	3632
B2_ExecRefiner	83.8	81.1	55.6	73.6	73.6	70.9	2889
B3_ValueMajorityVote	81.1	86.5	63.9	77.3	78.0	74.5	11416
B4_LLMJudge	83.8	86.5	58.3	76.4	77.0	72.7	11783
Argus-SQL	**83.8**	**83.8**	**58.3**	**75.5**	**76.4**	**75.5**	11532

trolled operator ladder in bundle `4200e72`, of which 880 cached `B0+`, `B1`, `B2`, and `Argus-SQL` evaluations were reused verbatim from the Iteration 4 checkpoint). First, on our $N = 22$ stratified verification slice, activating relational invariants I8–I10 (`LEFT_JOIN_FILTERED_RATIO`, `NON_KEY_ENTITY_GROUPING`, and `BOOLEAN_AVG_FOR_COUNT`) alongside 100% selected-table column retention doubled CHALLENGING EX (28.6% → 57.1%) and lifted overall Clean and Voice EX from 63.6% to 72.7%. Scaling to the 11-database $N = 110$ benchmark with multi-sentence ASR alignment, cardinality-hierarchy disambiguation, and invariants I11–I12 active (Table 4), **Argus-SQL** ranks **#1 of 7 on spoken `Voice-BIRD`**—achieving 75.5% **Voice EX** (83/110, 75.9 **R-VES**, 4.46 s median latency, 0.0% **Clean-to-Voice degradation**, and 6,836 Voice tokens—40.6% fewer than `B3`'s 11,514 and 42.8% fewer than `B4`'s 11,945) versus steelmanned value majority-vote (`B3`, 74.5%; one Argus win, zero losses), profiler-enriched full-DDL (`B0+`, 73.6%), LLM-judge (`B4`, 72.7%; five Argus wins vs. two losses), lexical-CoT and execution-refiner (`B1`/`B2`, 70.9%), and raw-DDL (`B0`, 69.1%; significant Voice gain, exact McNemar $p = 0.039$). The ladder also isolates the profiler: metadata comments alone lift `B0` to `B0+` by +1.8 Clean / +4.5 Voice EX points. On Clean text, **Argus-SQL** (75.5% EX, 83/110, 76.4 **R-VES**) trails steelmanned `B3` (77.3%, 85/110; $p = 0.625$) and `B4` (76.4%, 84/110; three Argus wins on QIDs 416, 533, and 744 vs. four losses on QIDs 24, 159, 371, and 743; $p = 1.000$), with no Clean pairwise gap reaching significance at $N = 110$ (exact McNemar $p \geq 0.27$). Against `B3`, Argus-SQL wins QID 533 via I11 `TIMESTAMP_BARE_DATE_COMPARISON` repair but loses three: on QID 24 a flagged `INTEGER_DIVISION_TRUNCATION` candidate projecting `satscores.sname` and its identical-result algebraic repair together outscore any single `frpm."School Name"` candidate; on QID 371 `formula_defensive` selects a `LEFT JOIN/COUNT(DISTINCT)` ratio whose denominator differs from BIRD's `INNER JOIN`; and on QID 743 `decomposed_cot` selects `INNER JOIN`s, whereas `B3`'s gold-matching `LEFT JOIN` keeps heroes with `NULL` foreign keys in the denominator. QID 24 exposes a concrete selection bug where keeping an identical-output repair alongside its flagged source double-counts that cluster above an unflagged single candidate, motivating repair-deduplicated cluster scoring alongside projection-aware tie-breaking.

5 Discussion, Limitations & Future Work

Scaling from SQLite to Terabyte Warehouses. While BIRD-SQL standardizes on SQLite, deploying **Argus-SQL** over multi-terabyte cloud analytical engines introduces partition-pruning and scan-cost constraints. Our read-only execution sandbox and R-VES evaluator naturally generalize to `EXPLAIN` dry-run cost estimation: before executing $K = 3$ candidate queries over billion-row fact tables, candidates missing partition-column predicates can be penalized as a thirteenth semantic invariant (`UNPARTITIONED_FULL_SCAN`).

Limitations. First, our deterministic intent canonicalizer handles lexical ASR disfluencies and common abbreviations but cannot resolve phonetic homophones of rare proper nouns without acoustic N-best hypotheses from the speech recognizer. Second, when a user's spoken query is genuinely underspecified (*e.g.*, *"show me top performers"* on a schema containing five distinct performance metrics), silent SQL selection is inferior to asking a brief clarifying voice question—motivating entropy-triggered interactive disambiguation over our equivalence clusters $\mathcal{C}_r$.

6 Conclusion

By systematically dissecting how existing NL2SQL paradigms fail across clean and spoken relational queries, we demonstrated that high accuracy need not require brute-force 20-sample ensembles. **Argus-SQL** establishes a principled accuracy–latency sweet spot—uniting voice canonicalization, FK-discounted Steiner schema linking, character 3-gram value grounding, adaptive batch synthesis, and zero-token algebraic dialect verification—offering a clean, extensible foundation for the next generation of voice-driven data interfaces.

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